

→ A system is INVERTIBLE if distinct I/p's leads to distinct o/p's.
If a system produces an identical o/p for two different I/p's, it doesn't have an inverse.

egs. of Non-invertible systems: $y(t) = x^2(t)$
 $y(t) = \frac{d}{dt} x(t)$
 $y(t) = \cos\{x(t)\}$

* L.T.I. SYSTEMS *

→ All C.T & D.T. signals can be represented as the sum of scaled & shifted unit impulses.

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k] \delta[n-k] \quad \left| \quad x(t) = \int_{-\infty}^{+\infty} x(\tau) \delta(t-\tau) d\tau \right.$$

Any DT signal is the sum of scaled and shifted unit impulses.

eg:-

→ $h[n]$ & $h(t)$ denotes the impulse response to be the o/p of an L.T.I. system when the i/p is a unit impulse $\delta(t)$ (or) $\delta[n]$

→ The response of an LTI system to an arbitrary i/p signal is the sum of its scaled & shifted impulse responses.

$$x[n] \rightarrow \boxed{\text{L.T.I. sys } h[n]} \rightarrow y[n] = x[n] * h[n]$$

$$x(t) \rightarrow \boxed{\text{L.T.I. System } h(t)} \rightarrow y(t) = x(t) * h(t)$$

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k] h[n-k]$$

Annotations for the discrete-time convolution sum:
 1. FLIP (on $h[n-k]$)
 2. SHIFT (on $h[n-k]$)
 3. MULTIPLY (on $x[k] h[n-k]$)
 4. SUM (on the summation)

$$y(t) = \int_{-\infty}^{+\infty} x(\tau) h(t-\tau) d\tau$$

Annotations for the continuous-time convolution integral:
 1. FLIP (on $h(t-\tau)$)
 2. SHIFT (on $h(t-\tau)$)
 3. MULTIPLY (on $x(\tau) h(t-\tau)$)
 4. INTEGRATE (on the integral)

→ CONVOLUTION involves finding the area under the product of one signal and a folded shifted version of another.

→ Convolution with an impulse simply produces a copy of the signal scaled & shifted by the size & location of the impulse.

$$a[n] * \delta[n-n_0] = a[n-n_0]$$

$$a(t) * \delta(t-t_0) = a(t-t_0)$$

→ DECONVOLUTION is the process of recovering the i/p signal when given the o/p of a system. It is the inverse of Convolution

→ Properties of CONVOLUTION:-

- Duration of o/p = Sum of the durations of $a(t)$ & $h(t)$

$$T_y = T_a + T_h$$

- Area of $y(t)$ equals the product of the areas of $a(t)$ & $h(t)$

$$A_y = A_a A_h$$

- $a^{(m)}(t) * h^{(n)}(t) = y^{(m+n)}(t)$ where m & n represents order of differentiation

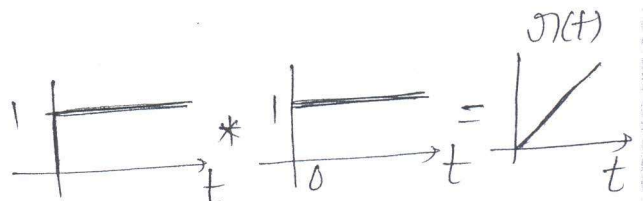
- $a(t-\alpha) * h(t-\beta) = y(t-\alpha-\beta)$

- Scaling $a(\alpha t) * h(\alpha t) = \frac{1}{|\alpha|} y(\alpha t)$

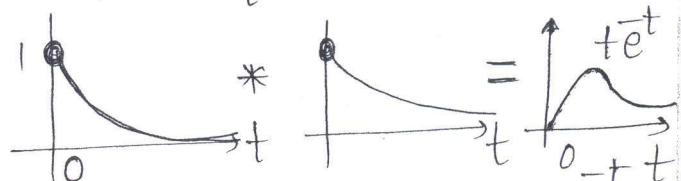
$$y(-t) = a(-t) * h(-t)$$

Useful Convolution results:

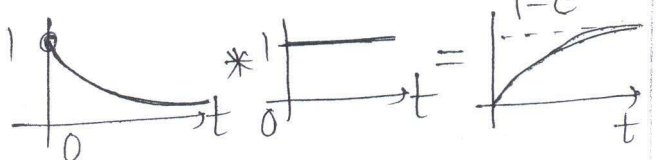
$$(1) \quad u(t) * u(t) = r(t) = t u(t)$$



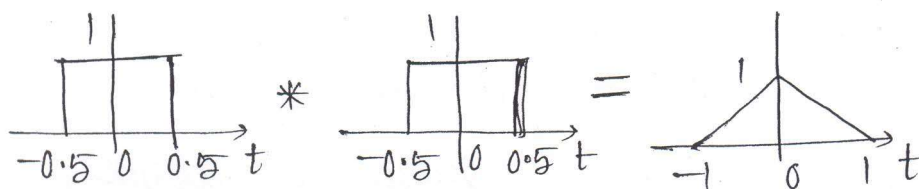
$$(2) \quad e^{-t} u(t) * e^{-t} u(t) = t e^{-t} u(t)$$



$$(3) \quad e^{-t} u(t) * u(t) = (1 - e^{-t}) u(t)$$



④



$$\text{rect}(t) * \text{rect}(t) = \text{tri}(t)$$

⑤ $u(n) * u(n) = (n+1)u(n)$

→ Impulse Response of N Interconnected L.T.I. Systems

In Cascade: Convolve the impulse responses

$$h_c(t) = h_1(t) * h_2(t) * \dots * h_N(t)$$

$$h_c(n) = h_1(n) * h_2(n) * \dots * h_N(n)$$

In Parallel: Add the impulse responses

$$h_p(t) = h_1(t) + h_2(t) + \dots + h_N(t)$$

$$h_p(n) = h_1(n) + h_2(n) + \dots + h_N(n)$$

→ Properties of L.T.I. Systems:-

Causal:-

$$h(t) = 0; t < 0$$

$$h(n) = 0; n < 0$$

Stable:-

$$\int_{-\infty}^{+\infty} |h(\tau)| d\tau < \infty$$

$$\sum_{k=-\infty}^{+\infty} |h(k)| < \infty$$

Memoryless:-

$$h(t) = K \delta(t)$$

$$h(n) = K \delta(n)$$

Invertible:-

$$h(t) * h_{inv}(t) = \delta(t)$$

$$h(n) * h_{inv}(n) = \delta(n)$$

eg:- ① $h(t) = e^{-3t} u(t-2) \rightarrow$ Causal & Stable

$h(n) = (1/2)^n u(n-3) \rightarrow$ Causal & Stable

② $h(t) = e^{3t} u(t+1) \rightarrow$ NonCausal & Unstable

$h(n) = 3^n u(n+2) \rightarrow$ NonCausal & Unstable

⑨